



Selected topics in Cryptography

Quiz 5A

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Problems

Read the following problem and solve it. Please write down the detail of all your calculations.

1. An RSA encryption scheme has the set-up parameters $p = 11$ and $q = 13$. The public key is $e = 7$

a) Decrypt the ciphertext $c = 37$ using CRT. (Value: 4 points)

b) Verify your result by encrypting the plaintext without using the CRT. (Value: 1 point)

Useful mathematics

Chinese Remainder Theorem

Given a ciphertext c , the decryption key d , the prime numbers for RSA p and q .

$$d_p = d \bmod p - 1$$

$$d_q = d \bmod q - 1$$

$$c_p = c^{d_p} \bmod p$$

$$c_q = c^{d_q} \bmod q$$

$$119 \bmod 143 = 119$$

$$119^2 \bmod 143 = 4$$

$$119^3 \bmod 143 = 4 \cdot 119 = 476 \bmod 143 = 47$$

$$119^4 \bmod 143 = 4 \cdot 4 = 16$$

$$119^5 \bmod 143 = 4 \cdot 47 = 188$$

$$119^6 \bmod 143 = 4 \cdot 4 \cdot 4 = 64$$

$$119^7 \bmod 143 = 47 \cdot 16 = 752 \bmod 143$$

$$= 37$$

$$120 = 7 \cdot 17 + 1$$

$$1 = 120 + 7(-17)$$

$$c = 37$$

$$p = 11$$

$$q = 13$$

$$n = 143$$

$$ph = 120$$

$$q = 7$$

$$d = -17 \cdot 120 = 120 - 17 = 103$$

$$d_p = 103 \bmod 10 = 3$$

$$d_q = 103 \bmod 12 = 7$$

$$c_p = 37^3 \bmod 11 = 9$$

$$c_q = 37^7 \bmod 13 = 2$$

$$k_{pub} = (143, 7) \quad k_{priv} = (11, 13,$$

$$q_i = 13^{-1} \bmod 11 = 6$$

$$p_i = 11^{-1} \bmod 13 = 6$$

$$C = 119^7 \bmod 143 = 37$$

$$m = (13 \cdot 6 \cdot 9 + 11 \cdot 6 \cdot 2) \bmod 143$$

$$= (702 + 132) \bmod 143$$

$$= 130 + 132 = 262 \bmod 143$$

$$= 119$$